Computing Economics

Course Objectives

- ✓ Why Study Economics?
- ✓ Definition of economics
- ✓ Difference between microeconomics and macroeconomics
- ✓ The economic problem: scarcity and choice
- ✓ The determinants of demand
- ✓ The determinants of supply
- ✓ Market equilibrium, surplus and shortage

Computational Economics

- Economics is the study of how we make decisions in resource-limited situations

- Computational economics is a form of economics which is constrained by the demands of computer science.

History

- The field started around 1999.
- Currently there are few dedicated conferences for computational economics.
- The field is growing very fast.

USES

1. Computer Science and Economics share the common ancestry from Mathematics.
2. Economics generate the problems from real life in terms of mathematical problems,
3. Computer Science is quite good at solving those mathematical problems.
4. With the rise of fields like data analytics, this relationship is only growing stronger.
5. Predicting the market trends, modeling the production, cost and efficiency, computer science is everywhere.
6. Economics can give you the theories to solve a problem. Computer Science actually solves it

Is The Study Of Economics Useful In Computer Engineering Fields?

- Perhaps if you choose to become a manager or start your own company, economics will be an asset in managing resources.
- If you are working on a project (usually as a team), knowing how much each resource costs will determine your objectives and your design process.
- one of the biggest complaints is that many employers complain that engineering graduates lack communication skills as well as interest in other fields.
- Perhaps for an engineer, it is important to have a working knowledge of different sectors because Computer Engineering is a very versatile field.

Why we computer science students are studying economics?

- ✓ To manage the production and development process of a organization.
- ✓To know about current knowledge of software market.
- ✓To know about industrial policy
- ✓To know about budget

Importance of economics

- ✓Computational Finance
- ✓Economic Forecasting
- ✓Online Trading and E-Commerce
- ✓Data Presentation
- ✓Mechanisms and market design
- ✓Applied market design
- ✓ Modern methods of production
- ✓ helps in proper budgeting

Importance of economics in Computer Science

- ✓Helps to idea of public demandable software build
- ✓Helps to idea of market price and software price
- ✓ Helps to idea of which software is essential according to our economy
- ✓Helps to idea of reducing software build cost
- ✓ Helps to idea of economic structure of a software and software firm
- ✓Helps to formulate budget
- ✓Overcoming Market Failure

Programming

- Needed to take ideas or business solutions and bringing them to reality.
- Small changes can be implemented in a short timeframe with a streamlined process
- Larger changes will require a software development methodology to guide and manage the process:
  - Systems development life cycle (SDLC)
  - Rapid application development (RAD)
  - Agile methodologies

Economics in Computing projects

- Rapid, sustained advances in computing & communications
- Hardware is seen as the catalyst, however software embodies new value added functions
- Despite this dependence on software its production is among the most complex and problematical aspects of modern technology development
- Lack of management in risk-return characteristics of software is a serious & difficult problem

Software Economics

- During software development, we are faced with numerous considerations, such as make-or-buy, how many options to build, which architecture to use, how much testing
- (prototyping, specifying) is enough, how much software to re-use, which requirements to add first, and how much time/staff are available.

Software Project Management Plan

- Software Project:
  - All *technical* and *managerial* activities required to deliver the deliverables to the client.
  - A software project has a specific duration, consumes resources and produces *work products*.
  - Management categories to complete a software project:
    - Tasks,
    - Activities,
    - Functions

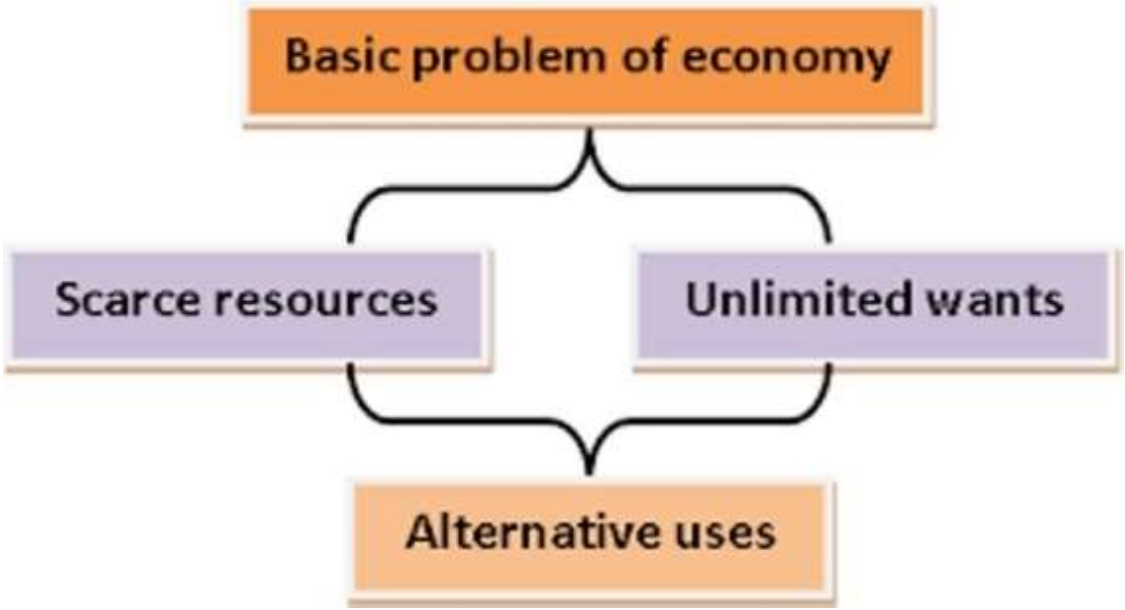
Software Project Management Plan

- The controlling document for a software project.
- Specifies the technical and managerial approaches to develop the software product.
- Economic document besides requirements analysis document: **Changes** in either may imply changes in the other document.
- Managing software projects is a part of computing economics.

The economic problem: scarcity and choice

The economic problem: scarcity and choice

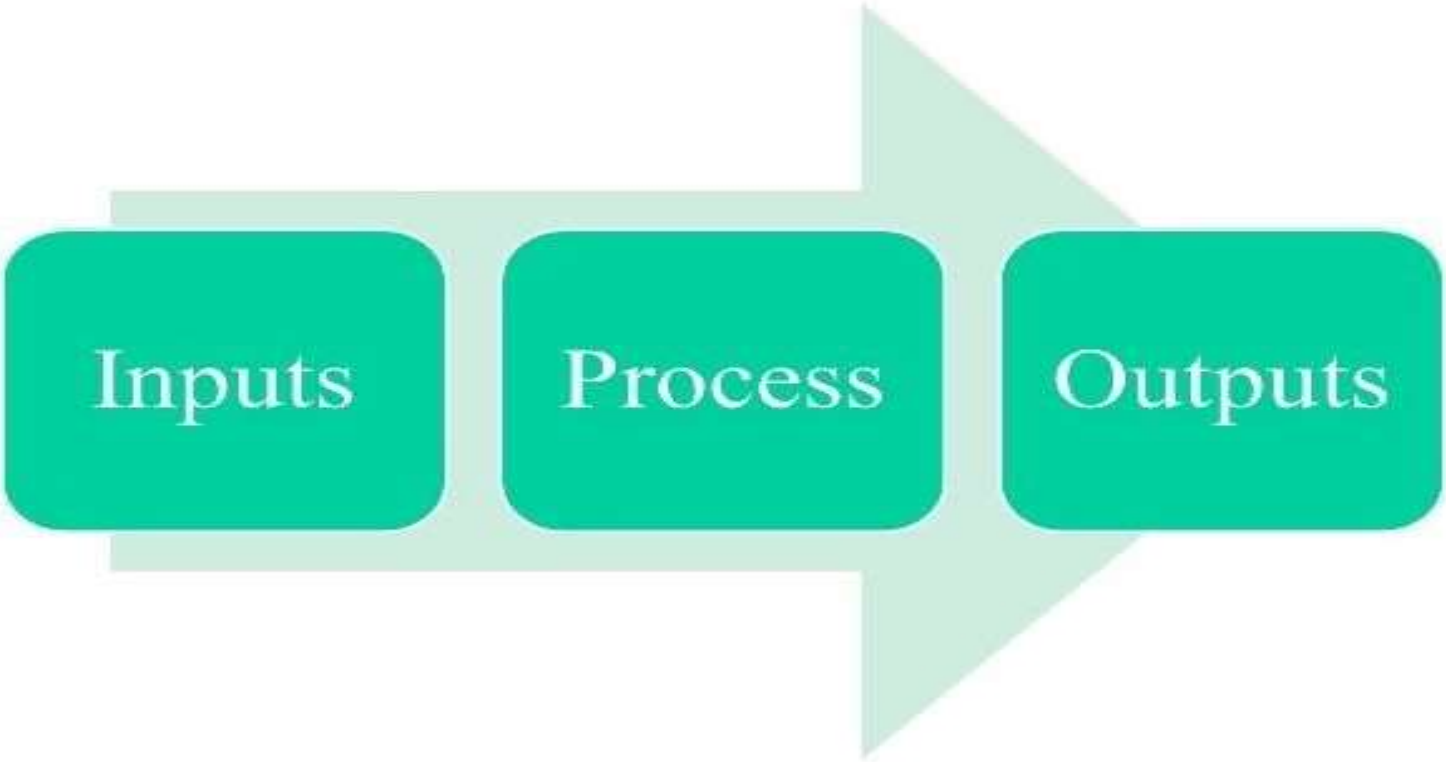
- Economy is derived from two Greek words which mean house and distribute.
- Economy was studied to understand the management of a household that later started being used to manage resources.
- The chart shows the basic problem of economy



Microeconomics vs Macroeconomics

Microeconomics	Macroeconomics
Is the study of individual economic units **Such as:** consumer, producer, market of a certain good.	Is the study of economics aggregates (the economy as a whole) **such as:** National income, Inflation, Unemployment, Economic growth, Gross domestic product (GDP).

Economic Operations



1 – Resources (inputs):

- Resources are factors of production used to produce goods & services.

Inputs

- **Resources consist of:**

a) **Natural resources:** such as land &raw materials (oil, minerals).

b) **Human resources:** labor both mental (doctors & lawyers) and physical (workers).

c) **Capital resources:** such as factories & machines used to produce goods & services.

d) **Entrepreneurs:** people who create businesses. They organize the factors of production.

2- Human wants (needs):

- **Human wants (needs):** consists of all goods (tangible) & services (intangible) that people would like to consume.

3- Production:

- **Production:** is the process of transforming inputs into outputs.

<u>Any economic system faces 3 questions</u> **<u>about production:</u>**

- What to produce & by what quantities?
- How to produce?
- How the production is distributed?

The economic problem

Problem of economics

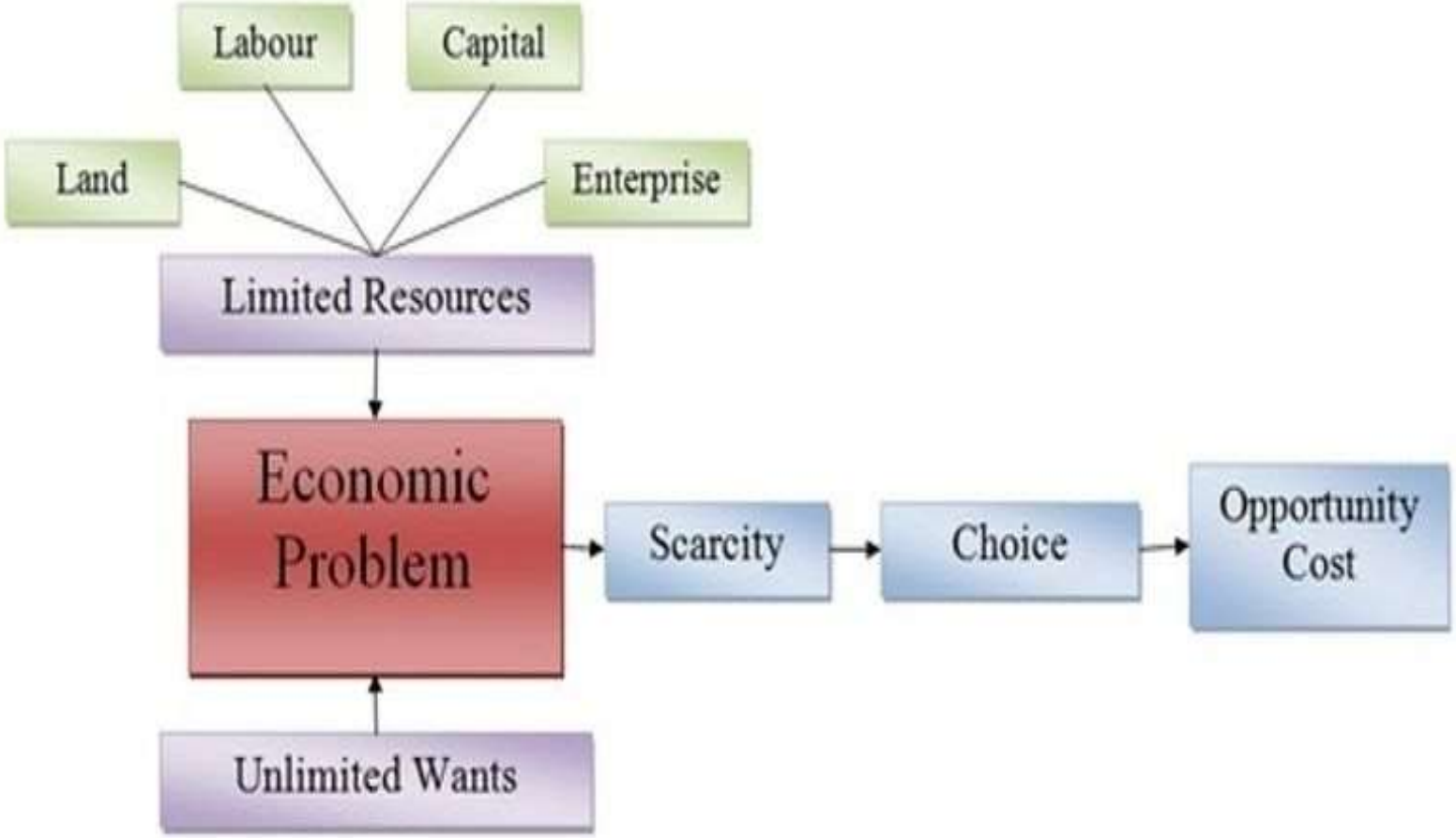
- Is a social science concerned with how individuals & society use or allocate their scarce (limited) resources to satisfy their unlimited human wants (needs).
- The basic problem of an economy deals with the needs and wants of a man being unlimited and the resources are scarce. The resources include the factors of production that are land, labour, capital and entrepreneurship
- **We distinguish between:**

1 Resources (inputs).

2 Human wants (needs).

3 Production.

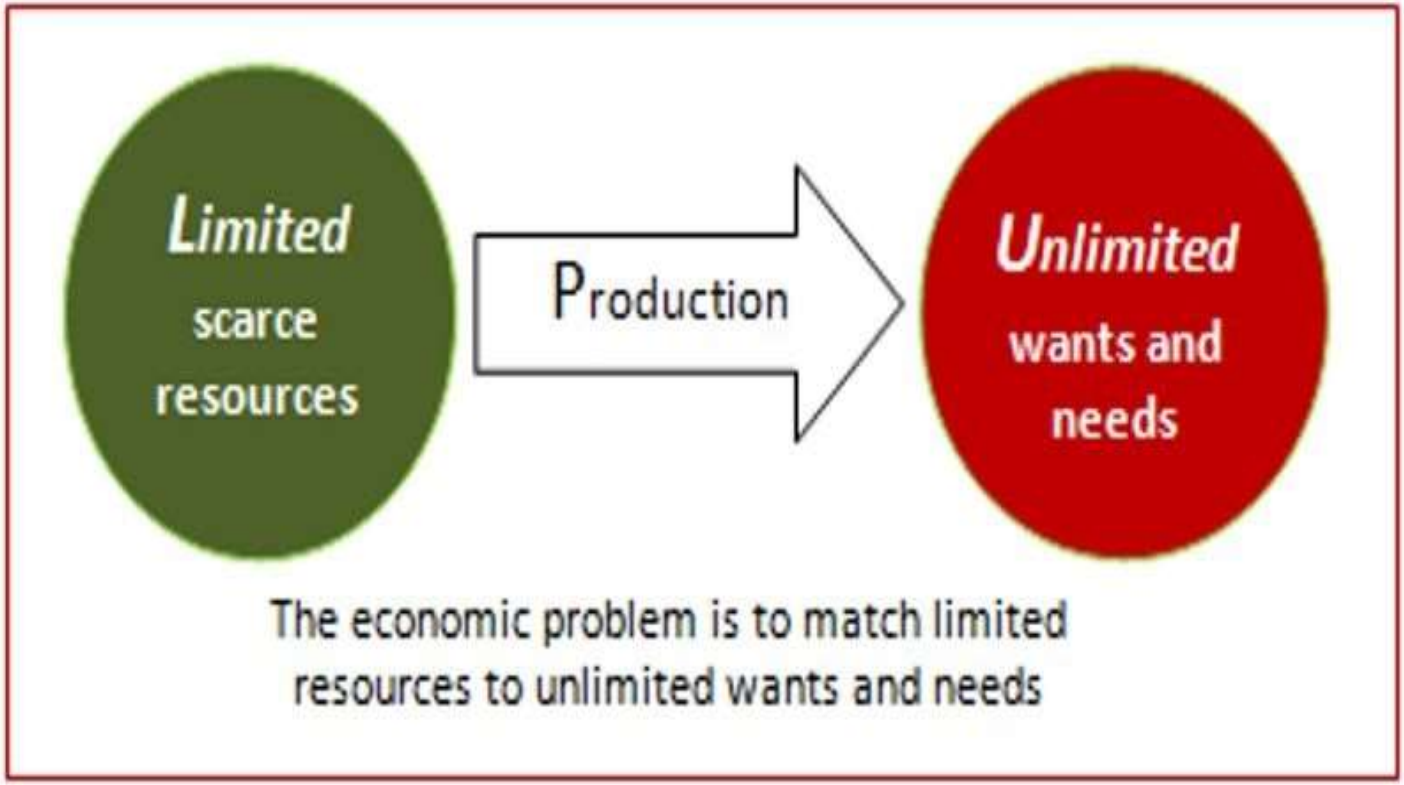
Economic Problem



Scarcity

- The purpose of production is to satisfy one's want but as the resources are limited, not enough output is available to fulfil every man's want.
- This explains that human wants are unlimited which are not fulfilled by the limited resources as stated by the Law of scarcity.

How law of scarcity arises?



The economic problem

ECONOMIC PROBLEM arises because of....

- *Scarce Resources*
- *Unlimited wants*

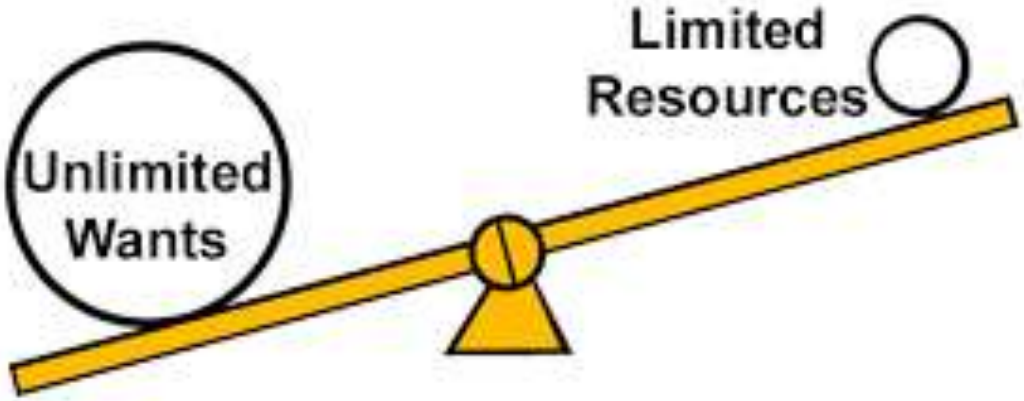
Economics studies the logic of choices made from among available possibilities

Problem 1: Resources are Scarce

- It means that resources are not enough to satisfy all human wants.
- All societies (poor & rich) face the problem of scarcity but at different degrees.
- scarcity requires that people must make a choice.
- Scarcity is the fundamental problem of the economy
- The problem of scarcity will never be solved and it will exists in every time, in the past or in the present or even in the future.

Problem 2: Human wants are unlimited

- Our wants are without limit and many of them cannot be satisfied.
- This is because there are simply not enough resources to make all the goods and services we want and need.



Problem 3: Scare resources have alternative uses

- Wants **unlimited** but resources are **scarce**.
- People, nations and the world must therefore **choose** how scarce resources are to be used.
- **Opportunity cost!**

Choice:

- As resources are scarce, all societies face the problem of deciding what to produce and the quantity to be produced.

Opportunity cost:

- Choice implies the existence of opportunity cost.
- The Opportunity cost of obtaining a unit of a certain good is measured by the number of the sacrificed units from another good.
- The opp. Cost is the best alternative you give up when you make any choice decision.

Opportunity Cost

The opportunity cost of a choice is the value of the best alternative forgone.

= The value of what you have given up.